

CS 307: Software Engineering

Homework #2

Due on March 5th, 2026 at 11:58pm
CS 307: Software Engineering Homework #2

Problem 1

This is a group homework problem. Your group will be same as the one that you are enrolled in for the team project in this course.

Find one open-source project that is closest to your team project. Reverse engineer the design and a few architectural/design patterns from that project.

Describe the above reverse engineered details in the following format. If applicable, provide pointers to project's directories/files/classes in your descriptions and/or diagrams.

1. Purpose (1 point)

- a. Briefly explain the chosen open-source project and its purpose.
- b. Provide the link to the open-source project repository.

2. Design Outline (2 points)

- a. Identify its components and describe the purpose of each component.
- b. Describe the interactions between components.
- c. Include at least one UML diagram that clearly shows high-level structure of the system.

3. Design Details (7 points)

- a. Include class level design (i.e., class diagrams) and be as detailed as you can.
- b. Describe the classes and interactions between the classes.
- c. Include sequence diagrams for different activities and/or activity diagrams and/or state diagrams (at least five UML diagrams).
- d. In your descriptions and/or diagrams, point to the source file and line numbers if applicable, so that the grader can have a look.

4. Design Patterns (4 points)

- a. Provide textual description of the patterns along with their diagrams (at least five patterns).
- b. In your descriptions and/or diagrams, point to the source file and line numbers if applicable, so that the grader can have a look.

5. Design Issues (1 point)

- a. Identify one functional design issue and one non-functional design issue.
- b. Each design issue requires a descriptive title, the solution option chosen by the

project, and the alternative solution that you can think of which will either degrade/improve the design.

Submission

Please make sure to include team number and the names of all your team members. You are expected to follow the organization of the design documentation with a right information in the right section and appropriate pointers to the project repository content in your descriptions and/or diagrams.

A homework assignment will be created on Gradescope (only one submission per group). You are expected to submit a document in PDF format.

Violation of the above instructions will result in a reduction in score.

Open Source Project links:

- <https://github.com/zahraaMeky/ai-travel-planner-app>
- <https://github.com/barika001/ai-trip-planner>
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